

Mars Outpost Mission Proposal

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Scope Summary

Need: To find out if there is life on Mars/how to sustain life on Mars

Goal: To collect data on the environment and climate of Mars for find a solution to sustaining life on Mars

Objective: Gathering data on the geology, climate, environment, etc. to see how Mars has changed over the years to determine if there is life on Mars and how to sustain life on Mars.

Mission: Send rovers and satellites to collect data about Mars for a year and help further the mission of sustaining life on Mars.

Mars Outpost Mission Proposal

As the expansion to new horizons continues, preparation and precautions must be taken in consideration. As the exploration of Mars continues, more information about the planet is learned. Humans have discovered Mars ' environment and climate through the rovers going all over the planet. One of the reasons for this exploration is to discover what Mars has to offer us. Since Earth and Mars are similar in many ways, why not allow human beings to live on the red planet. The initial discoveries were just a start. The discoveries of a new livable planet could lead to new resources that could benefit humanity in the long run. This proposal will go over the importance of the exploration of Mars through determining if life on Mars is possible through the climate and geology of Mars. Thus, this will better prepare for living on Mars and other planets. It will also go over the requirements and restraints for this mission.

Mission Name

The name of this mission will be Seeds. The reasoning for this name is to represent the overall goal of this mission. The main mission is to live on Mars and sustain life on Mars. Seeds usually grow into a tree and when it grows up, it spreads its seeds to other areas. Similar to what humanity is doing with this mission. NASA is spreading its seeds to new planets so that humanity will be able to live on new planets and learn from the planet. Seeds also spark hope for new life for humanity. The hope is that by living on Mars, NASA will be able to find discoveries that will not only help the community living on Mars, but also the people still on Earth. Seeds represent a cycle for humanity to grow on new planets and find new discoveries.

Mission Objective

To understand the importance of this Mission, there needs to be reasons why this mission needs to be completed. As stated before, the goal of this mission is to find a way to sustain life on Mars. With this, learning about the geology could provide information on the age of rocks and what specifically happened on the planet. The composition could point to water which is a sign that life might be on that planet (Goals, 2020). Rovers can analyze the rock and send that data to scientists to research on. Comparing the climate to Earth gives astronauts an understanding of the weather that goes on. It will give an understanding of what the weather was like in the past. Comparing it to Earth gives it a baseline of how it became a cold desert (Goals, 2020). Rovers could take wind and weather readings.. Finding out if Mars can support life is crucial to finding out if Mars holds new life forms of its own and if Mars can support human life. NASA would need to search for the main factors that all life needs which is water. They will need to search for water if there is a chance to find life (Goals, 2020). They can also search for fossils and such to see if life existed before. Rovers can search for water and dig for fossils. This satisfies the goal of determining how life will ever arise on Mars. Because of this mystery, the mission needs to happen to satisfy the goal.

Timeline

Year	Events
2023	● Planning of the Mars base begins ● NASA organizes the materials needed ● Astronauts will begin training for living and working on the Mars base

2024	● NASA will send up materials to the Mars near Eberswalde ● Construction begins with robots and astronauts building the base ● Communications are set up between Earth and the Base ● Solar Panels are set for collecting power for the base
2025	● Living area with food, water, and air sources are developed with reusable methods ● Shuttles and other traversal vehicles are created ● Shuttle Launch/Landing pads are created for crew ● Lab will be created for experiments and robotics
2026	● Base is deemed safe and stable to live and work in ● Astronauts finish training for operating Mars Base

	● Important medical equipment will be sent incase of an emergency
2027	● Construction on a shuttle to carry a 20 man crew begins ● Materials and supplies are being sent
2032	● Shuttle is now completed and double checked for safety precautions ● All 20 Astronauts are sent up with personal belongings such as hygiene in June.
2033	● All astronauts arrive at the base and are sent to their designated workplace and living quarters
2034-2048	● After a year of confirming the base is safe to live in, the base is fully operational in May 25, 2034 ● Experiments and research begins
2048	● Crew will begin packing up ● Shuttle will begin to set course to

	Earth
2049-2050	• Crew lands on Earth

Mission Subject

The mission will aid humanity in the search for life beyond Earth. The mission will include rovers that will investigate the climate and soil composition of Mars as well as search for fossils and water. NASA will send off meanderers and satellites to Mars to build the exploration done in the world. The rover will send images of the terrain on Mars and any signs of life to Earth. In order to determine its age and composition, the soil will be collected and broken down. The environment will be estimated with a mix of satellites and wanderers. The satellite will follow the climate and the meanderers will discuss that information with one another and to researchers on The planet. This data will be used to learn more about the planet's history and how it got to be what it is today. The objectives are fulfilled and it will be the base point for additional examination and disclosures.

Mission Users

The findings and research carried out on Mars will be beneficial to mission users. Governments would want to communicate with and make peace with this new life if it were discovered on Mars. Knowledge would be shared and a new culture would be learned. Scientists would like to investigate the new life there and try to determine how it differs from humans or is similar to them. The geology and environment of Mars would be of interest to scientists looking for fossils. The evolution of Earth and other planets can be better understood by studying their geology (Why go to Mars, n.d.). The terrain of Mars can be studied more thoroughly by scientists. Additionally, the information gathered by the satellites and rovers will contribute to

the enhancement of NASA's rovers for future Mars exploration, possibly by humans. It will also transmit any information regarding biohazards that must be considered prior to travel there (Why go to Mars, n.d.). This mission has huge advantages.

Operational Concepts

This mission's operational ideas will make it easier to keep the mission organized. The two rovers and one satellite for Mars exploration will be built in March 2024. The launch's preparations and planning will begin after and are scheduled for June 2024. According to Trip to Mars, n.d., the journey to Mars will take approximately seven months and span 300 million miles. The rovers and satellites will begin their work in January 2025. The rocks and geology will be observed by the first rover. It will collect rock samples to scan and send to Earth, as well as take photographs of the terrain. The second one will collect weather and wind data. Additionally, it will search the surroundings for water and signs of life. It will communicate with the satellite and function with climate. This will go on for a year. Some materials needed are the materials needed to build the rover and the satellite, the rocket to launch them, and the communications network needed to connect the rovers and satellite with the base on Earth.

Landing Site

For the best possible efficiency and ease of entry into Mars, the landing site is crucial. The landing site needs to be in a location where the rovers can easily connect to the satellite and where rocks and soil can be collected. Because it is built to handle rocky and desert terrain, the first rover can land at any location. It was designed to observe the terrain and collect rock samples. The rover ought to go somewhere high in order to gather climate data and information on locations near water sources. The streambeds in Eberswalde, a crater located at 24°S 33°W, are higher than the surrounding area. It is additionally accepted that there was likewise water in

this hole. Additionally, there is wind-eroded water-deposited sediment (Eight potential landing sites, n.d.). This is a good place to get weather readings and a closer look at how this planet could have had life. The location of the landing is crucial to the mission's outcome.



Crew Responsibilities

With a crew as large with 20 members, it is important to have roles. As a group as large as them, there needs to be specific roles of each member to do otherwise, the expedition will result in chaos and miscommunication. To prevent that miscommunication, there needs to be a Mission leader and Assistant leader to keep the crew organized. This crew leader must know the basics of every task and be qualified with exceptional leadership skills. They will also handle communication and inform the crew on Earth about their progress on Mars. From there each crew member will be assigned a role based on how well their expertise in the task. There are members who would perform the experiments and collect data based on the climate and geography, there are members who fix rovers and any machinery in the base. There are also members who serve as medical advisors and help any crew member who feels ill. With these roles, it ensures that the base will be maintained and the safety of the crew and base will be good.

Mission and System Requirements

The mission and system must have the specific requirements for each goal to be fulfilled. The mission requirements are important as if they are not fulfilled, the mission goal will not be fulfilled. The requirements for this mission are the materials and supplies that are needed to keep the crew and base operational. The rovers and satellites that are required to gather data and samples from Mars are crucial for experiments and data to fulfill the goals of the mission. The satellites also serve a purpose of establishing communication between Earth and Mars. All scheduled activities such as arrival and departure will be on time and punctual. Once the base is set up and samples are collected, experiments to help humans survive on Mars will begin immediately. With these requirements fulfilled, the mission will go on as planned.

Constraints

Even though the mission is planned out, there are some constraints that occur. The main constraints that NASA has put is that the launch must occur before 2040, the base must be a phase III base, which is composed of 10-40 people, and the crew must come back to Earth by 2050. The timing of the launch would not be much of a problem as the start of the project would begin immediately as it would take at least 5 years to build a shuttle (Welcome to how things fly, n.d.). NASA will get help from other organizations such as ESA and government funding to keep up the pace. To speed things up, as soon as the rovers are built, they will be sent to Mars immediately to build the base while the astronauts are training. Another constraint is getting the rocket into space and getting the trajectory correct. The design might not be able to get the shuttle to Mars. The next constraint is the budget and materials. Rovers and satellites are expensive with the Mars Exploration Rover costing \$1.08 billion (Cost of the Mars exploration rovers, n.d.). There needs to be a budget so that the rovers and satellites will work properly without overspending on the budget. The location and timing needs to be good since the location

of the launch will help get the rocket to Mars faster and the mission itself has no delays. To get the crew home before 2050, the crew will pack everything in 2048 and double check everything before they launch to do home. The trip will take 7-8 months for them to go home and they are expected to arrive on Earth in 2049-2050 depending on delays and unexpected situations. Even though the constraints are difficult, there are solutions to them.

Risk Assessment

With a plan, there are both rewards and many risks. A good way to reduce the likelihood of risks occurring is through preparation. Space exploration necessitates the development of a sound plan that identifies and addresses potential dangers and offers a response in the event of their occurrence. Radiation is the most significant threat to consider. Astronauts face a significant health risk from radiation. Radiation can separate DNA atoms and harm cells, which can cause medical conditions (How does radiation influence the human body in space?, 2006). Diarrhea and vomiting are two examples of mild health issues, while damage to the central nervous system, an increased risk of cancer, or even death, are more severe. Mutated genes even run the risk of transmitting the health issues to subsequent generations (How does radiation affect the human body in space?, 2006). More examinations have found that attractive fields can divert modest quantities of radiation (Space Faring The Radiation Challenge, 2008). An electron can be knocked out of orbit by ionizing radiation, causing enormous damage (Why Space Radiation Matters?, 2017). Radiation affects not only humans, but also technology like computers and materials that could affect the space shuttle if exposed to it. To combat this, astronauts can shield themselves from radiation with barriers or other means. In order to guarantee the safety of the astronauts and the space shuttle, the shield will replicate the global magnetic field of the Earth

but will be ten times stronger. On Mars' surface or wherever there isn't a strong enough magnetic field to shield against radiation, this will be used. In the event of a breach in the shield, the astronauts will receive instruction on how to sustain an ill astronaut. Precautions will guarantee that the astronauts will be aware of how radiation can affect the mission and how to deal with being exposed to it, despite the fact that radiation poses a threat to the safety of the astronauts and the technology of the space shuttle.

Food, water, shelter, and air are the four most important things that living things need to survive. Space travelers can't make due without oxygen and will endure medical issue like queasiness and power outages if breathing low convergences of oxygen. (Space Station Breathing Easy, n.d.). Technologies for air recycling and filtering have been utilized on the International Space Station for some time. The technology delivers electricity to the water, separating the atoms into oxygen and hydrogen. Why does the International Space Station never run out of gas?, 2022). Future shuttles need to use this technology for long space missions because breathing air means fewer health problems. Because of this technology, not only will there be fewer oxygen tanks to carry, but the shuttle may also have a longer lifespan. Astronauts can go further into space while meeting all four basic needs of a human being using recycling technology.

The crew is able to decide whether a mission is successful or unsuccessful. There are numerous sizes and shapes of astronauts. To determine who is qualified to go, the selection must take place. The astronaut must be able to work with others and adapt to a new environment physically and mentally. They will be more of a hindrance to the team if they do not meet those requirements. The crew will undergo a self-evaluation test for these risks to determine their suitability for space travel. They will be evaluated regarding their performance in space, their understanding of shuttle technology and piloting, and their medical suitability for space travel.

Individual and group training will be provided to an astronaut once they have demonstrated their ability to live and work in space. Maintaining the space shuttle and working professionally require teamwork. The work will never be finished if the team is unable to cooperate. Therefore, a group of astronauts will be assigned to collaborate on numerous tasks before the astronauts are launched into space. They will be assessed on how quick they can finish the responsibility, how they collaborate with one another, the moves toward tackle the issue, and who assumes responsibility and goes about as the pioneer. Although each astronaut plays a significant role, the crew as a whole must cooperate to complete the mission.

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Appendix

Space exploration for many years has been a goal. Long-term flight into space has been difficult because of our limitations as humans. However, humankind is the reason space exploration has reached its current state. We need to push ourselves to go above and beyond and conquer the risks that come our way. Despite their apparent impossibility, these three risks can be mitigated. Radiation presents the first risk. Space radiation can cause numerous health issues for humans. However, there is a workaround for this. Ionizing radiation can be avoided by using a magnetic shield, which can safeguard both humans and technology. Because of this, astronauts will be able to travel further into space without having to worry about getting sick from radiation. Breathable air is another danger, as oxygen is necessary for humans to breathe clean air or they will get sick. Additional oxygen tanks and oxygen recycling through electrolysis, which converts water molecules into hydrogen and oxygen, will be the solution to this problem. Because of this, astronauts won't have to worry about air. Even though there are a lot of risks, there have been measures taken to reduce those chances to a minimum. The selection process for a team with the necessary physical and mental attributes is in progress. After the group sessions and interviews are over, crew training will begin. Training will teach basic astronaut skills, how to use and fix technology, how to communicate, and what to do in an emergency. There has been great advancement in finding competitors that could finish the mission at any point as well as manage what is going on that space investigation tosses at them. Space exploration is one risk that is worth taking, so knowing these risks will help us learn how to avoid them. Space provides new information and concepts that humans could not have imagined. With new technology, this could make life better for people and assist others in curing diseases. We still haven't seen a lot of the universe. With the work we have placed into it, space investigation is an unquestionable

requirement. However, we must refrain from rushing into action and examining obstacles to be overcome.